

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. This cycle's overall qualitative score set (2026-09-05_draft2) is a FULL WEEKLY refresh, COMMITTED BY ALEX SEPTEMBER 5, 2026 (weekly refreshes are categorically excluded from auto-stamp, and are Alex's own periodic review checkpoint): three qualitative pillars (CLARITY Act, Tokenization RWA, Iran War Risk) held flat, and Fed & Macro re-confirmed its Sept 5 re-rate down on a hot August jobs report and higher FOMC hike-odds pricing — see that pillar report. This quant pillar's own output is unaffected by qualitative status and enters the master blend the same way regardless.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through September 5, 2026 (BTC \$79,646.11) · Pipeline re-run & verified September 5, 2026 (FULL WEEKLY refresh, 2026-09-05_draft2) · Not Financial Advice

The Scores — September 5 Model Output

Computed September 5, 2026 (previous computed run September 2, 2026; no new quant refresh ran between then and this cycle, so these are the same computed figures re-verified) — same standard as every qualitative pillar. This cycle's mapping ticks up marginally at the front/mid end and holds everywhere else as the cycle-month clock advances: 1mo 57→59 (up on the day's data refresh), 3mo 47→48 (up a touch), 6mo 59→60 (up a touch), 1yr 68→68 (flat), 3yr 69→69 (flat) — the weekly bar that closed above the 200-week MA on Aug 23 remains confirmed; no new weekly/monthly/quarterly bar closed this cycle, so completed-bar composites are unchanged. This cycle's overall qualitative score set (2026-09-05_draft2) is a FULL WEEKLY refresh, COMMITTED BY ALEX SEPTEMBER 5, 2026 (Fed & Macro re-confirmed its re-rate down; the other three qualitative pillars held flat); this quant pillar is computed fresh every run regardless and carries no draft/committed status of its own.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	59	0.75	8	+1.08	Mild Bull — the weekly close above the 200-week MA remains confirmed (weekly composite as-is +38.6, Slight bullish, 6-of-7); the 1mo cycle-analog holds 2-of-3 positive, median +3.6%
3 months	48	0.70	8	-0.22	Neutral — still the softest window, 1-of-3 analog positive, median -5.3%
6 months	60	0.55	8	+0.88	Mild Bull — unanimous 3-of-3 positive, median +43.0%
1 year	68	0.60	8	+1.73	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3), median +80.8%
3 years	69	0.50	8	+1.52	Bull — value + cycle (36-month analog 3-of-3 positive, median +1351.8%)

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P - 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term still genuinely bullish, not just repairing (weekly composite +27.7/100, Slight bullish, agreement 4-of-7 — the weekly bar that closed above the 200-week MA on Aug 23 at \$77,716.48 remains a confirmed completed-bar signal); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, median +70.4%, and price sits 45% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (842 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: -50% from the all-time high just weeks ago, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price has since recovered to \$79,646.11, roughly flat over Sept 3-5 after a ~2% jobs-day pullback (\$81,265.00 → \$79,676.60) on the hot August jobs report — still +36.1%

off the June 30 low; the tape moving on its own schedule is itself information: that move came from real-world macro news, not from anything in the TA model itself.)

The model: the short-term tape remains genuinely confirmed, not just repaired — the weekly composite holds Slight bullish (+38.6, agreement 6-of-7) and the quarterly composite holds Slight bullish (+26.6, agreement 4-of-7); the monthly composite is essentially flat (+7.3, Neutral) with agreement at 2-of-7. No new weekly, monthly, or quarterly bar closed this cycle, so completed-bar composites are unchanged: weekly +31.2, monthly +1.8, quarterly +7.7. The weekly bar dated 2026-08-23 that closed above the 200-week MA (\$77,716.48) remains a confirmed completed-bar signal — Repair Signal #1 stays confirmed, not a lean. The 1-month window firmed to Mild Bull (P 59, up from 57) as the 1-month cycle analog holds 2-of-3 positive (median +3.6%); 3 months holds Neutral (P 48, up a touch). The 6-month window firmed to Mild Bull (P 60) with the analog still unanimous (3-of-3 positive, median +43.0%). The long read holds its direction (1yr flat at 68, 3yr flat at 69): every prior cycle at this exact month was higher 12 months later (median +80.8%), price is 43% below fair value (\$139,417), and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the tape) — and the tape has held "confirmed bullish" at the near end even as the qualitative side of this system logged a fresh macro-driven cut. The model still reads three — tape (confirmed at the weekly level), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.55, past the average-bottom date), and valuation (43% below the power-law fair line, with price sitting well above the –1σ "cheap" band). All three clocks point the same direction on the medium-to-long horizon — the near-term clock has stayed caught up to the other two, and notably firmed rather than reacted negatively to this cycle's jobs-report-driven macro news.

The Facts — The Tape

*Computed September 5, 2026 from daily data (BTC \$79,646.11, daily close as of September 5). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite holds at **+31.2** (confirmed since the weekly bar dated 2026-08-23 closed above the 200-week MA); monthly +1.8 and quarterly +7.7 are also unchanged — no new bar closed this cycle.*

- **Weekly composite +38.6/100 (Slight bullish), agreement 6-of-7 indicators** — firmed as the cycle-month clock advanced and price recovered from the Sept 4 jobs-day dip. Price is above the 20-week (\$70,136, +13.6%) and well above the **200-week average (\$64,866, +22.8%)**; price sits just under the 50-week (\$80,331, –0.9%). RSI 58.4; MACD line negative (–1,482) with histogram positive (+2,400) — momentum repair continuing; %B 0.82. Price (\$79,646.11) remains above the falling weekly resistance trendline (\$76,216) that it broke above earlier this cycle.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) remains CONFIRMED.** The weekly bar dated 2026-08-23 closed at \$77,716.48, +20.9% above its 200-week average (\$64,286.55) — a genuine completed-bar close above the line, not a still-forming daily read. The completed-bar weekly composite holds at **+31.2** since that close — the clearest single confirmation this pillar has logged this cycle.
- **Monthly composite +7.3/100 (Neutral), agreement 2-of-7** — essentially flat as September's still-forming bar absorbs the week's price action. Below the 20-month average (\$87,083, –8.5%), above the 50-month (\$62,178, +28.1%); RSI 51.5; monthly MACD line +407 with histogram –3,657 — still negative; %B 0.39. Completed-bar monthly composite (August's closed bar) holds at +1.8, unchanged.
- **Quarterly composite +26.6/100 (Slight bullish), agreement 4-of-7** — firmed with the cycle-month clock. Comfortably above the 20-quarter average (\$58,204, +36.8%) and the 50-quarter (\$28,758, +176.9%); RSI 57.6, MACD line +16,316 with histogram –362. Completed-bar quarterly composite is +7.7, unchanged (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price (\$79,646.11) sits between the old \$74,434 support (–6.5%, 1 touch) and fresh overhead resistance at \$81,685 (+2.6%, 2x touch) — the nearest test on the board — then \$89,164 (+11.9%, 1x) and \$98,090 (+23.2%, 2x). Further below, \$59,930 (–24.8%, 1x) and \$57,735 (–27.5%, 1x). The falling weekly support

trendline sits at \$43,728; the falling resistance trendline, at \$76,216, remains broken to the upside, price now holding comfortably above it.

- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has since built a **+36.1% bounce to \$79,646.11**. **\$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.



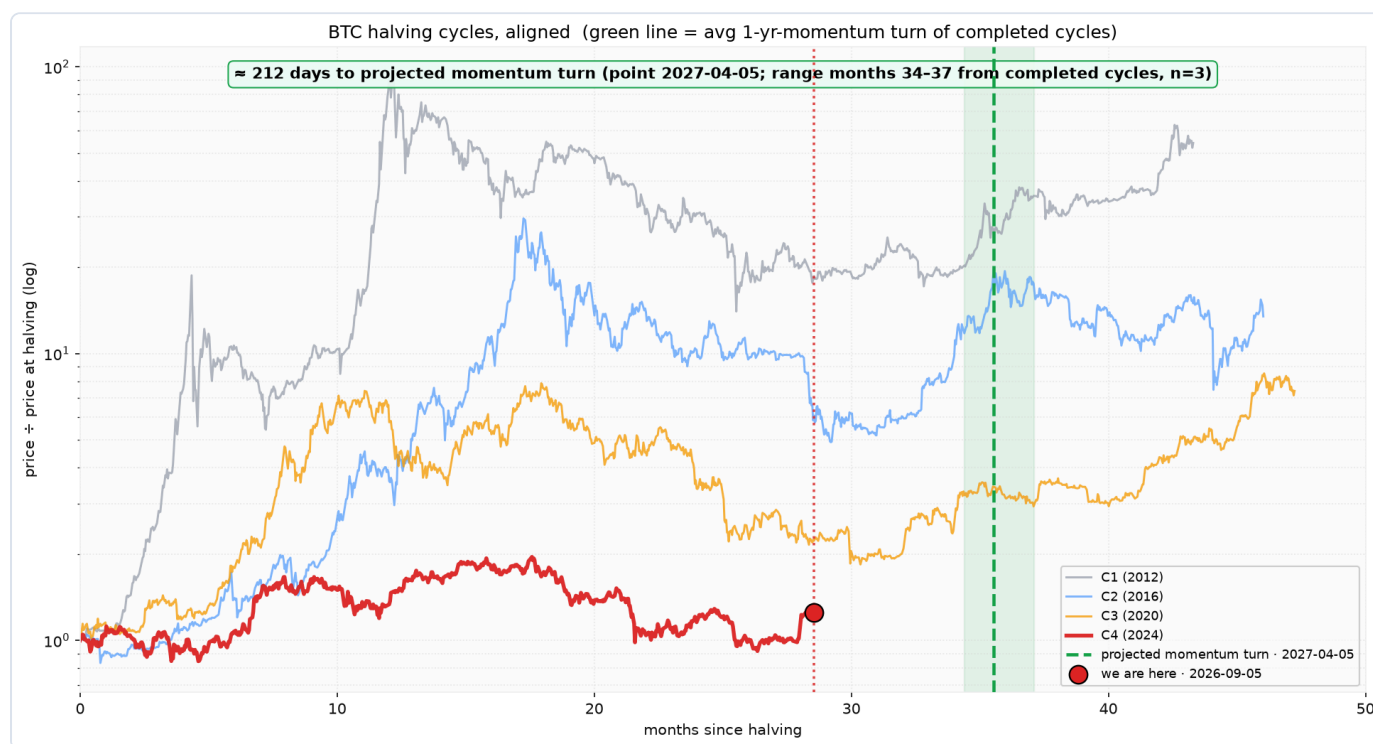
Weekly tape, scored +38.6/100 (Slight bullish) with agreement 6 of 7 indicators: the weekly close above the 200-week MA remains confirmed on a completed bar, price holds comfortably above the falling resistance trendline, MACD histogram remains positive. Machine-annotated from the same data that feeds the scores.

The Facts — The Cycle Clock & The Value Floor

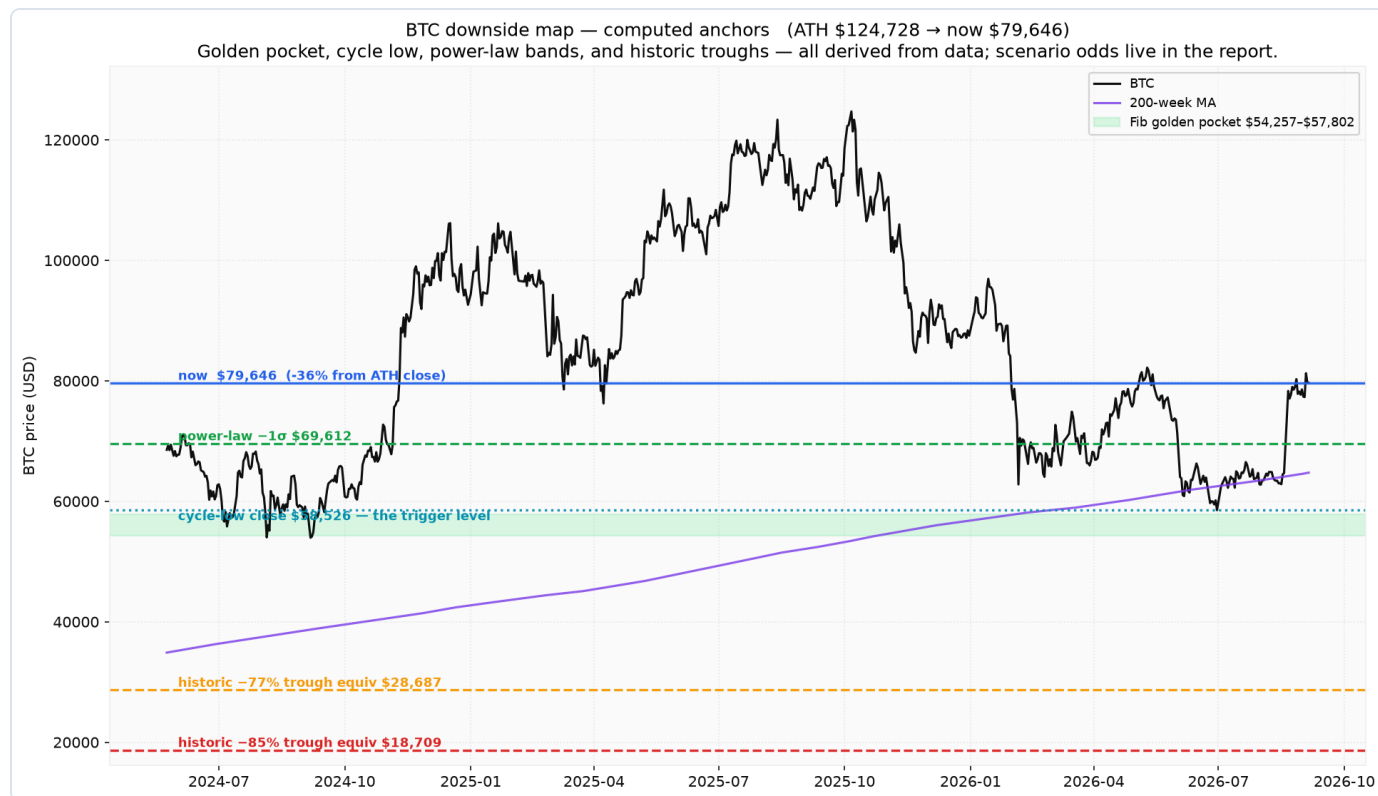
The two slower clocks. Same data, longer lenses.

- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30, \$58,526, –53% from the top). **We are at month 28.55 — past the average-bottom date.**
- ★ **The cycle analogs are the spine of the long-window scores.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **2-of-3** (median +3.6%) · next 3 months positive **1-of-3** (median –5.3%) · next 6 months positive **3-of-3** (median +43.0%) · next 12 months positive **3-of-3** (median +80.8%) · next 36 months positive **3-of-3** (median +1351.8%). *Fragility disclosure: the medians and agreement counts move as the cycle-month clock advances — treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~28 that window reaches month ~64 — it contains the NEXT halving's rally.*
- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5) → still projects to ~April 2027** (80% window Jan–Jul 2027) for this cycle.

- **The power-law floor:** a log-log fit over the full price series puts fair value at **\$139,417** — price is **-43%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value to \$129,226 / \$134,176 / \$117,572 / \$144,442 respectively (fair value ranges roughly \$117K–\$144K depending on fit window) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price (\$79,646.11) sits above the **-1 σ band (\$69,612)** — out of the "cheap" zone (the -2 σ band sits at \$34,758, far below). Every halving-era cycle trough was -77% to -85% from ATH (\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went ~-93% (~\$8.7K equivalent), excluded from the range but in the data; this cycle's low-so-far was -53% from the ATH.
- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to -39% average 12-month drawdowns vs -23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23, unchanged)* — used as a tilt, not a score input.
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 28.45, past the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$79,646.11: the 20-week average (~\$70.1K) is well behind us — price sits **+13.6%** above it; the 50-week average (\$80,331) is back overhead as the immediate test, now just **0.9%** away — while the falling weekly trendline resistance (\$76,216) remains broken to the upside, held comfortably. A retest of the \$124,728 ATH is **+56.6%**; the power-law fair line is **+75.0%**. The median 12-month analog from this cycle-month is now **+80.8%** (still shy of a full retest of the prior ATH but well above last cycle's +47.0%); the median 36-month analog is +1351.8% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the near-term signal remains confirmed, and firmed further this cycle even after the Sept 4 jobs-day pullback — the near quarter is still the softest window (P 48), but the 6-month-and-beyond case remains unanimous (3-of-3) and essentially unmoved by a single week's macro news.

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.

- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$64,286	Weekly close above the 200-week MA	CONFIRMED — the weekly bar dated 2026-08-23 closed at \$77,716.48, +20.9% above this level; completed-bar weekly composite jumped -29.5 → +27.6 accordingly
~\$80,331	Weekly close above the 50-week MA	Price sits -0.9% below it — the next real test; a confirmed weekly close above it would extend trend repair further — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) has now passed (we're at month 28.55)	Price sits well off the June 30 low (+36.1%); a higher low + improving monthly momentum after this date = basing confirmed — August's completed monthly bar (+1.8) is an early such sign
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	-60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: COMMITTED (FULL WEEKLY REFRESH, 2026-09-05_draft2, STAMPED BY ALEX) · THIS PILLAR: COMPUTED SEPTEMBER 5, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 59 / 48 / 60 / 68 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read firmed this cycle: the weekly bar that closed above the 200-week MA on Aug 23 stays a confirmed completed-bar signal, and the completed-bar weekly composite holds at +31.2 while the as-is weekly composite firmed to +38.6 (6-of-7). The 1-month window firmed to Mild Bull (P 59, from 57) as the 1-month cycle analog holds 2-of-3 positive. The 6-month window firmed to Mild Bull (P 60) as the cycle-analog for that horizon remains unanimous (3-of-3 positive, median +43.0%). The 3-month window ticks up marginally (P 48, from 47) — still the softest window in the set. The long read holds its direction (1yr flat at 68, 3yr flat at 69) as the cycle-month clock advanced to 28.55 — the direction, not the level, is the signal.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: this cycle's overall qualitative score set (2026-09-05_draft2) is a FULL WEEKLY refresh, COMMITTED BY ALEX SEPTEMBER 5, 2026 (Fed & Macro re-confirmed its re-rate down on a hot August jobs report and higher FOMC hike-odds pricing; the other three qualitative pillars held flat). This pillar's own computed output is unaffected by that process and enters the master blend the same way it does every cycle.

What Changed Since Last Check

Pipeline re-run September 5, 2026 (previous computed run September 2, 2026).

- **Price: \$76,581.44 (Sept 2) → \$81,265.00 (Sept 3) → \$79,676.60 (Sept 4) → \$79,646.11 (Sept 5).** The Sept 3→4 pullback (~2%) was a macro-driven move (hot August jobs report), not a TA-driven one. June 30's \$58,526 remains the cycle low; price is now +36.1% off it.
- ★ **Repair signal #1 remains CONFIRMED: the weekly bar dated 2026-08-23 closed at \$77,716.48, +20.9% above the 200-week average (\$64,286.55).** This is a genuine completed-bar close, not a still-forming daily read — the strongest single confirmation this pillar has logged this cycle.
- **No new weekly, monthly, or quarterly bar closed this cycle** — completed-bar composites are unchanged: weekly +31.2 (confirmed since the Aug 23 weekly close above the 200-week MA), monthly +1.8 (August's closed bar), quarterly +7.7 (current bar still forming). On the current (forming) bars, versus Sept 2: weekly +27.7→+38.6 (agreement 4-of-7→6-of-7, still Slight bullish), monthly +1.4→+7.3 (agreement holds at 2-of-7, still essentially flat), quarterly +23.6→+26.6 (agreement 4-of-7, still Slight bullish) — all three firming with the price recovery.
- **Model P vs Sept 2: 57/47/59/68/69 → 59/48/60/68/69.** 1mo +2 and 6mo +1 on the day's data refresh; 3mo +1; 1yr flat; 3yr flat.
- **Cycle analogs held their agreement counts this cycle (2/3, 1/3, 3/3, 3/3, 3/3)** as the cycle-month clock advanced to 28.55, with medians ticking (1mo +2.2%→+3.6%, 3mo -8.8%→-5.3%, 6mo +45.3%→+43.0%, 1yr +70.4%→+80.8%, 3yr +1272.5%→+1351.8%) — flagged per this report's standing n=3 fragility disclosure, not treated as new structural information.
- **Power-law fair value: \$139,151 (Sept 2) → \$139,417 (Sept 5)** (price -45% → -43% vs. fair, the narrower gap driven by the higher price) — normal drift as new daily bars extend the fit.
- **Data refreshed:** daily price data through September 5, 2026 (data/btc_usd_daily.csv, via the GitHub Actions fetch workflow); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a

score input).

- **Context:** this cycle's overall qualitative score set (2026-09-05_draft2) is a FULL WEEKLY refresh, COMMITTED BY ALEX SEPTEMBER 5, 2026 — Fed & Macro re-confirmed its re-rate down on a hot August jobs report and higher FOMC hike-odds pricing; Iran War Risk, CLARITY Act, and Tokenization RWA held flat. This pillar's own computed output is independent of that process and enters the master blend the same way regardless of the qualitative set's status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,835 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Sept 5 (completed-bar composites: weekly +31.2, confirmed by the Aug 23 weekly close above the 200-week MA / monthly +1.8, August's bar having closed / quarterly +7.7, current bar still forming — none changed this cycle since no new bar closed) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · $n=3$ on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.